Emmanuel Mombo-Dingba

Computer Engineer | AI & Machine Learning Engineer

Currently AI Engineer @ HADRON S.A.

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SCAN PORTFOLIO
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PROFILE

Computer Engineer (ENSPY) and AI Engineer at HADRON S.A. with proven expertise in Artificial Intelligence, Machine Learning, Multi-Agent Systems, and enterprise Large Language Models (LLMs). Experienced in designing and integrating AI-driven solutions into production systems (ERP, microservices, real-time event-driven pipelines). Solid mathematical foundation (Linear Algebra, Calculus, Probability, Statistics, Optimization) combined with advanced proficiency in Python, FastAPI, Docker, and modern ML/DL frameworks.

EDUCATION

National Advanced School of Engineering (ENSPY) — Yaounde, Cameroon
Engineering Degree in Computer Engineering — **2021 – 2026**

University of Ngaoundere — Ngaoundere, Cameroon
Undergraduate Studies in Pure Mathematics — **2019 – 2021**

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, TypeScript, Java, Bash
AI & Machine Learning: Machine Learning, Deep Learning, NLP, Large Language Models (LLMs), RAG, Multi-Agent Systems
Deep Learning: Neural Networks, CNNs, RNNs, LSTMs, Transformers, Attention Mechanisms
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, LangGraph, Pandas, NumPy, OpenCV
Data & Visualization: PostgreSQL, MySQL, Redis, Redis Streams, Kafka, Matplotlib, Seaborn
Software & Deployment: Git, GitHub, GitLab, Docker, Linux, REST APIs, Microservices, Celery, Jupyter, Google Colab

PROFESSIONAL EXPERIENCE

AI Engineer **[CURRENT POSITION]** — **Feb 2026 – Present**
HADRON S.A. — Yaounde, Cameroon

- Contributed to the design and implementation of an intelligent logistics platform integrated with an enterprise ERP system.
- Designed and integrated AI-driven components to support the automation and optimization of logistics workflows.
- Worked on a Multi-Agent System (MAS) architecture coordinating intelligent services and automating business processes.
- Integrated a conversational AI assistant based on Large Language Models (LLMs) and NLP for natural-language interaction and planning assistance.
- Contributed to the software architecture of AI services using Python, PostgreSQL, Redis Streams and microservices.

Technologies: Python, LangChain, LangGraph, LLMs, NLP, PostgreSQL, Redis Streams, Docker, Microservices, Celery

Software Engineering Intern **Jul 2025 – Sep 2025**

National Advanced School of Engineering (ENSPY) Yaounde, Cameroon

- Developed a real-time communication platform supporting messaging and peer-to-peer video communication.
- Implemented real-time communication using Socket.IO and WebRTC protocols with sub-100ms latency.
- Contributed to backend architecture, REST APIs and real-time data synchronization.
- Collaborated in an Agile development environment through sprint planning, code reviews and technical discussions.

Technologies: React.js, Node.js, Socket.IO, WebRTC, REST APIs, Flutter

SELECTED AI PROJECTS

Autonomous Agentic ERP Bridge & Logistics Assistant **HADRON S.A.**

- Event-driven autonomous system automating ERP decision-making (Inventory & Logistics) via intelligent reasoning agents.
- Engineered Redis Streams streaming pipeline, LangGraph contextual reasoning, and Pydantic deterministic safety guardrails.
- Maintains sub-300ms decision latency on 95% of operational events with a 99.2% automated validation rate.

Technologies: Python, FastAPI, LangGraph, Redis Streams, PostgreSQL, Docker, Celery

Diabetes Prediction Using Machine Learning Open Source / ML Lab

- Developed and compared multiple classification models including Logistic Regression, Support Vector Machines and Random Forest.
- Performed exploratory data analysis, feature analysis and preprocessing including missing-value handling, outlier analysis and feature scaling.
- Evaluated model performance using Accuracy, Precision, Recall, F1-score and ROC-AUC metrics.

Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

AI & MACHINE LEARNING KNOWLEDGE

Machine Learning: Supervised and Unsupervised Learning, Classification, Regression, Model Evaluation, Feature Engineering

Deep Learning: Neural Networks, CNNs, RNNs, LSTMs, Transformers, Spatial Attention

Natural Language Processing: Text Processing, Language Models, Large Language Models, Prompt Engineering, Retrieval-Augmented Generation

Mathematical Foundation: Linear Algebra, Calculus, Probability, Statistics and Optimization

ADDITIONAL AI/ML TRAINING

Self-Directed Study Using Stanford Online Resources **2023 – Present**

- Machine Learning concepts inspired by CS229 (Supervised/Unsupervised Learning, Kernel Methods, Learning Theory).
- Deep Learning & Computer Vision concepts inspired by CS231n (Convolutional Networks, Spatial Features, Object Detection).
- Natural Language Processing concepts inspired by CS224n (Word Embeddings, Transformers, Attention Mechanisms, LLMs).

LANGUAGES & CORE INTERESTS

French: Native English: Professional working proficiency

Research Interests: AI for Healthcare Diagnostics, Agricultural Optimization in Africa, Scalable ML Systems